

Batch: A3 Roll No.: 16010324044
Experiment / assignment / tutorial No. 1
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Plot p-n junction diode IV characteristics and determine its parameters

AIM: To determine IV characteristics of p-n junction diode.

1. **OUTCOME:** Student will be able to Analyze diode parameters

Theory:

Biasing of PN junction Diode:

Forward bias operation: The P-N junction supports uni-directional current flow. If +ve terminal of the input supply is connected to p-side and -ve terminal is connected to the n-side, then diode is said to be forward biased condition. In this condition the height of the potential barrier at the junction is lowered by an amount equal to given forward biasing voltage. Both the holes from p-side and electrons from n-side cross the junction simultaneously thereby decreasing the depleted region. This constitutes a forward current (majority carrier movement – diffusion current). Assuming current flowing through the diode to be very large, the diode can be approximated as short circuited switch. Diode offers a very small resistance called forward resistance (few ohms)

Reverse bias operation: If negative terminal of the input supply is connected to p-side and +ve terminal is connected to n-side then the diode is said to be reverse biased. In this condition an amount equal to reverse biasing voltage increases the height of the potential barrier at the junction. Both the holes on P-side and electrons on N-side tend to move away from the junction thereby increasing the depleted region. However the process cannot continue indefinitely, thus a small current called reverse saturation current continues to flow in the diode. This current is negligible; the diode can be approximated as an open circuited switch it offers a very high resistance called reverse resistance (few Kiloohms).

Static Resistance: The opposition offered by a diode to the direct current flowing forward bias condition is known as its DC forward resistance or Static Resistance. It is measured by taking the ratio of DC voltage across the diode to the DC current flowing through it at an operating point.

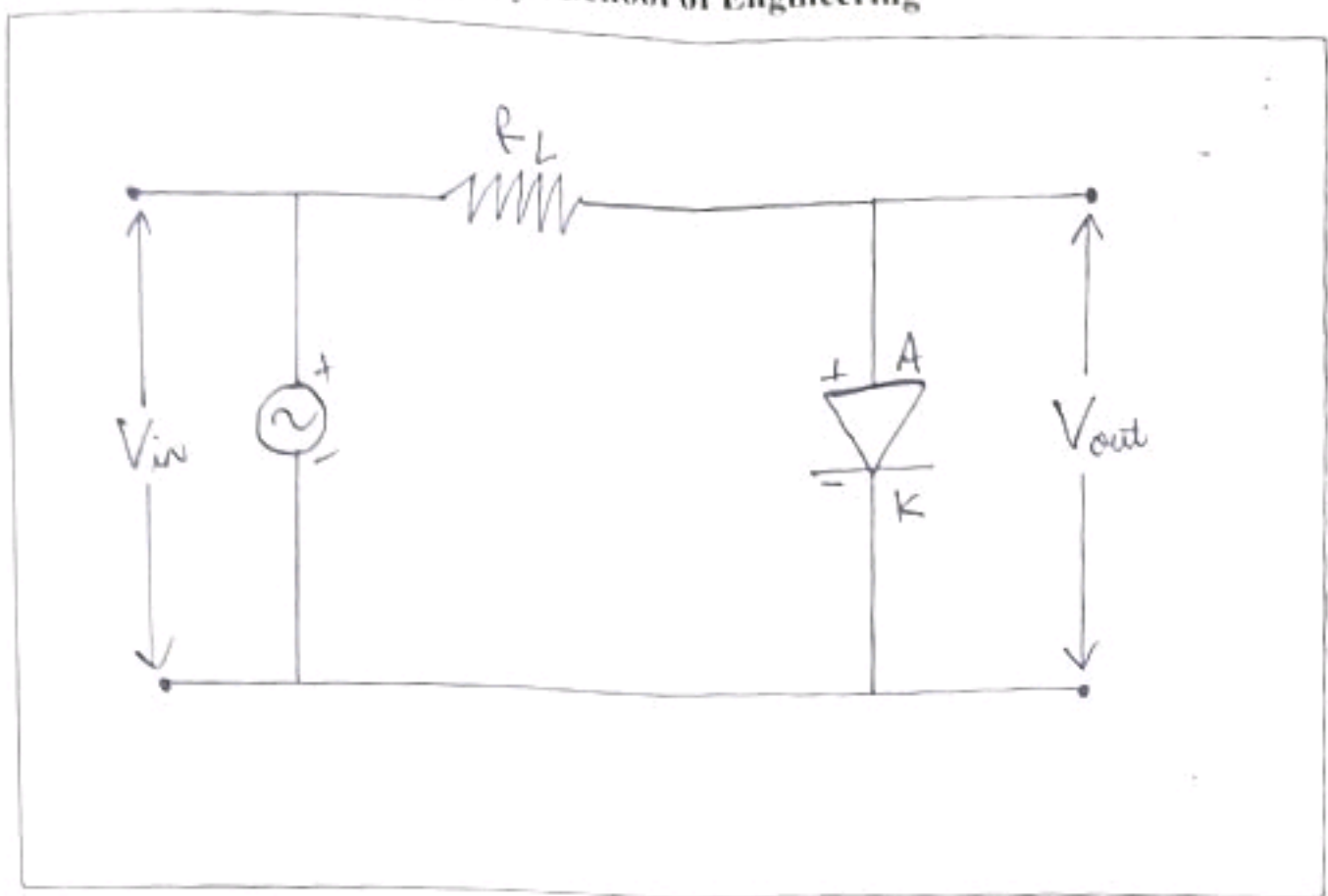
Dynamic Resistance: The opposition offered by a diode to the changing current flow in forward bias condition is known as its AC Forward Resistance. It is measured by a ratio of change in voltage across the diode to the resulting change in current through it for an operating point P.

Average Resistance: Same as dynamic resistance but measured between extremities.

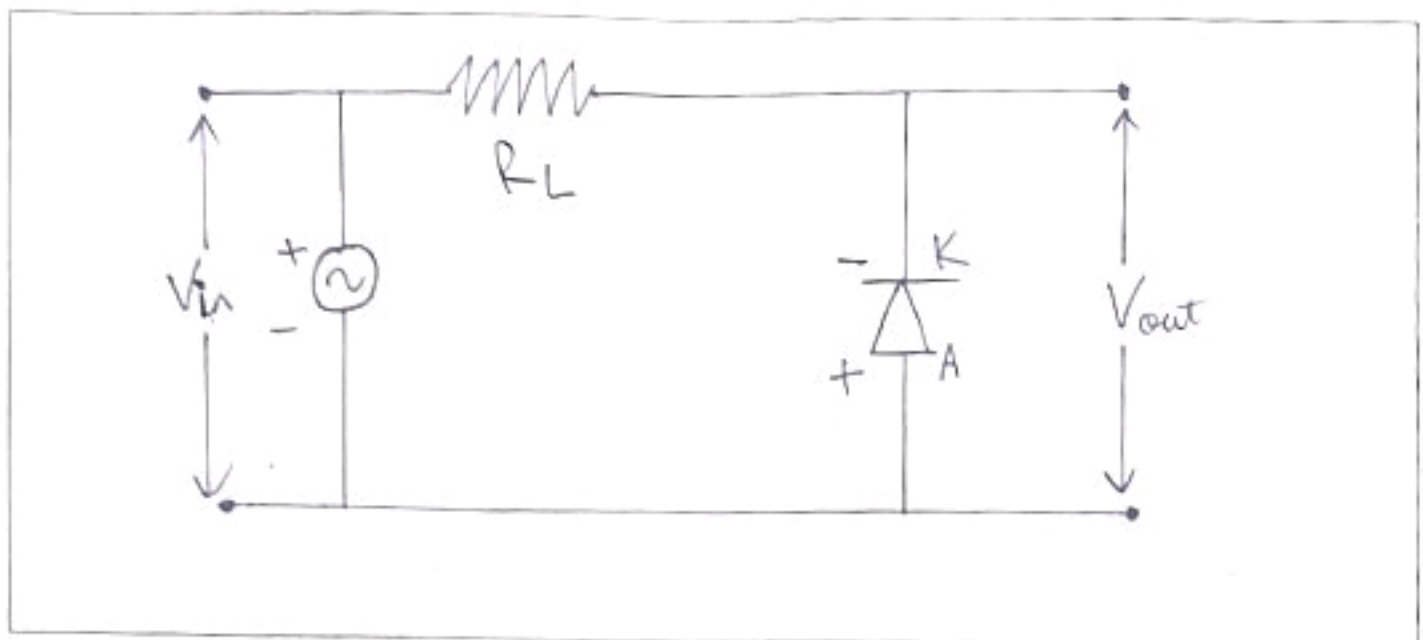
Procedure:

1. Implement the setup to determine IV characteristics of p-n junction diode on breadboard.
2. Note down the observations and Plot the IV characteristics.
3. Determine diode forward and reverse resistance from the characteristics.

Circuit diagram (Forward characteristics)



Circuit Diagram (Reverse characteristics)



Calculations from Graph:

Cut-in Voltage $V = 0.7 \text{ V}$

forward Resistance = $10^3 \Omega$

Reverse resistance = $10^7 \Omega$

Observation Table:

① Forward Characteristics:

<u>I (mA)</u>	<u>V (Volt)</u>
41	0.1
0.00526	0.2
0.0237	0.3
0.0636	0.4
0.1052	0.5
0.1052	0.6
0.250	0.7
0.378	0.8
0.965	1
2.424	2
3.429	3
4.414	4
5.271	5

② Reverse Characteristics:

<u>I (μA)</u>	<u>V (Volt)</u>
-0.2	-10V
-0.3	-20
-0.35	-30
-0.37	-40
-0.4	-50
-0.43	-60
-0.46	-70
-1.85	-80
-10.6	-81
-32.9	-82
-63.3	-83
-94.6	-84
-135.8	-85
-165.7	-86
-196.7	-87
-215.7	-88
-237.3	-89

IV Characteristics

Forward Characteristics	
I (mA)	V (Volts)
0.0	0.02 μ
0.1	0.04 μ
0.2	0.5 μ
0.3	4.95 μ
0.4	21.82 μ
0.5	73.4 μ
0.6	98.2 μ
0.7	132.05 μ
0.8	234.5 μ
0.9	282 μ
1	344.4 μ
2	1.055 m
3	2.45 1.677 m
4	3.45 m
5	4.46 m

Reverse Characteristics	
I (mA)	V (Volts)
0.0	0.5
0.1	0.45
0.2	0.4
0.3	0.35
0.4	0.31
0.5	0.28
0.6	0.25
0.7	0.20
0.8	0.18
0.9	0.15
1	0.14
2	0.05
3	0.01
4	0.00
5	0.0

→ PTD

IV Characteristics (Graph)

Calculations:-

① Forward resistance

$$\text{Slope} \rightarrow \frac{(4.5 - 3.5) \times 10^{-3}}{5.4} = \frac{10^{-3}}{5.4} = 0.001 \Omega$$

$$R = 1 \text{ m}\Omega$$

② Reverse resistance

$$\text{Slope} \rightarrow \frac{(-1.6 - (-1.4)) \times 10^{-6}}{-4 - (-2)} = \frac{(-0.2) \times 10^{-6}}{-2}$$

$$= 0.1 \times 10^{-6} \mu\Omega$$

$$R = \frac{1}{0.1 \times 10^{-6}} = 10 \times 10^6 = 10^7 \Omega$$

Results:

Q. 1. Explain the diode current equation.

We learnt about diode current,
Forward Bias & Reverse Bias of
Diode.

Q1 Conclusion:

→ The diode current equation
is given by Shockley diode
eqn.

Signature of Faculty with
date

(a) i.e. $I_D = I_S [e^{\frac{V}{nV_T}} - 1]$ $V_T = \frac{KT}{q} = 26 \text{ mV (at RT)}$

Diode current reverse saturation I_S is ideality factor depending on diode material and quality. $(1 < n < 2)$

(b) When diode is FB. ($V > 0$), the exponential term dominates, $I \uparrow$ rapidly.

(c) When diode is RB. ($V < 0$), the current is approx I_S , very small in magnitude. $\therefore$ at low voltages (close to 0), current is nearly zero the exponential term is small.

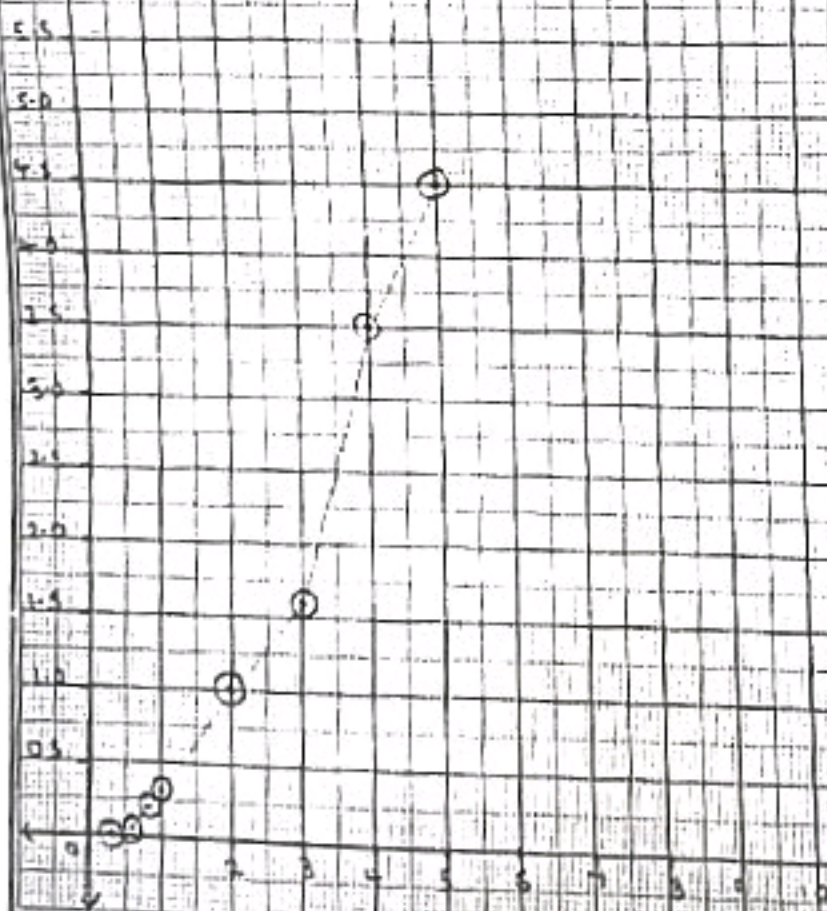
(d) At higher forward Voltages (eg. 0.7V for silicon), the exponential term grows fast, causing a steep current rise.

(e) This explains the diode's non-linear I-V char.

(f) The eqn. is fundamental in understanding diodes, rectifiers, LEDs and Transistors.

(a)

On X-axis, time in
on Y-axis, time in 0.5 sec



Avaash Reddy

On X-axis, $I_{cm} = 1V$
On Y-axis, $I_{cm} = 0.2mA$

